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Review

Depression and cardiovascular diseases

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ABSTRACT

Depression is a well-known risk factor for adverse cardiovascular outcomes in patients with cardiovascular diseases. The prevalence of depression in patients with cardiovascular diseases has been reported to be approximately 20 %. A two-step depression screening protocol using the 2-item Patient Health Questionnaire (PHQ-2) and the 9-item Patient Health Questionnaire (PHQ-9) is recommended for patients with cardiovascular diseases. Cardiovascular diseases and depression share a common pathology, including increased activity of the sympathetic nervous system, hyperactivity of hypothalamic-pituitary-adrenal axis, and inflammation. Psychosocial and environmental factors are also associated with depression and cardiovascular outcomes. Randomized controlled trials of antidepressant treatment for patients with depression and cardiovascular diseases have shown no advantage regarding cardiovascular outcomes. However, improvement in depressive symptoms, regardless of the method, may lead to a reduction in subsequent cardiovascular events. A collaborative approach between cardiologists and psychiatrists is recommended to manage depression in patients with cardiovascular diseases. Future research should identify more specific targets for treating patients with cardiovascular diseases, involve collaboration with professionals across fields, and establish community support systems.

Introduction

Depression is a risk factor for adverse cardiovascular outcomes in patients with various cardiovascular diseases [1–8]. This mood disorder is closely associated with physiological function, elevates the risk of suicide, and is a risk factor for worse prognoses of physical illnesses. Various risk factors for cardiovascular diseases have been discovered over the years, and the management of these risk factors has improved patient prognosis. However, despite advances in the treatment of cardiovascular diseases, progress in improving patient prognosis is restricted by limitations. Depression is also an independent risk factor for cardiovascular diseases that should not be overlooked by cardiologists. In this review, we will discuss the relationship between depression and cardiovascular diseases and provide recommendations to approach this association in the future.

Prevalence of depression in patients with cardiovascular disease

The prevalence of depression in patients with coronary artery disease (CAD) is approximately 20 %, with some reports suggesting that it may reach as high as 30 % to 40 % in patients with heart failure (HF) [6,9–13]. The prevalence of depression in patients with cardiovascular diseases has doubled over the past several decades [14]. Moreover, patients with myocardial infarction or HF have a threefold higher risk of depression than the general population [2,8].

A study that analyzed the National Health Interview Survey (NHIS) data of 30,801 U.S. adults reported that the 12-month prevalence rates of major depression disorder (MDD) were 9.3 % in patients with CAD, 8.0 % in patients with hypertension, and 7.9 % in patients with HF; these rates are all higher than that in patients without chronic disease (4.8 %) [15]. Moullec et al. reported that 6 % of 750 outpatients referred to undergo an exercise trial for CAD had MDD [16]. In Japan, a prospective observational study reported that 22 % of inpatients with cardiovascular diseases had depression, and 5.6 % of outpatients also had depresssion [17,18].

In patients with HF, a meta-analysis of 27 clinical studies reported that the frequency of depression was approximately 20 %, and this frequency was similar in both inpatients and outpatients [10]. Moreover, patients in higher classes (i.e. more severe symptoms) according to the New York Heart Association (NYHA) classification of cardiac function had higher frequencies of depression. Although nonischemic HF is

Abbreviations: AHA, American Heart Association; BDI, Beck Depression Inventory; CAD, Coronary artery disease; CBT, Cognitive-behavioral therapy; CRP, C-reactive protein; HADS, Hospital Anxiety and Depression Scale; HF, Heart Failure; HPA, Hypothalamic-pituitary-adrenal; ICD, Implantable cardioverter defibrillator; IL, Interleukin; MDD, Major depression disorder; PHQ, Patient Health Questionnaire; SADHART-CHF, Sertraline Against Depression and Heart Disease in Chronic Heart Failure; SAM, Sympathetic-adreno-medullar; SDS, Zung Self-Rating Depression Scale.

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more common in Japanese patients than in Western patients, it accounts for about approximately 20 % of cases [19,20].

Of patients who received implantable cardioverter defibrillators (ICDs), 24–33 % had depression [21]. Similarly, we reported that approximately 30 % of 90 patients who received ICDs had depression according to the Zung Self-Rating Depression Scale (SDS), which has a threshold score of ≥60 [22]. Bilge et al. reported that depression scores on the Hospital Anxiety and Depression Scale (HADS) were higher in patients who had ICDs for longer periods (>5 years) and in those who had received recent shock therapy (within the past 6 months) [23]. Friedmann et al. also reported that Beck Depression Inventory (BDI) scores were higher in patients who had ICDs for more than a year than in patients who had ICDs for less than a year [24].

Screening and diagnosis of depression

Most reports on the prevalence of depression associated with cardiovascular and other diseases have used screening tools such as questionnaires to assess depression. Although this method is useful for a simple survey of a large number of subjects, these questionnaire scores are not a diagnostic criterion for depression and should be considered only as an assessment of depressive symptoms. The diagnosis of depression and assessment of the severity of depressive symptoms require an interview with a psychiatrist. In particular, the diagnosis and specialized treatment of MDD and related disorders, including bipolar disorder, require a psychiatrist assessment.

To date, the BDI, SDS, HADS, and the Centre for Epidemiologic Studies Depression Scale (CES-D) have been used to assess depressive symptoms. The American Heart Association (AHA) has recommended a two-step depression screening protocol using the 2-item Patient Health Questionnaire (PHQ-2) and the 9-item Patient Health Questionnaire (PHQ-9), especially for patients with CAD [25]. The PHQ-9 items are consistent with the American Psychiatric Association's Diagnostic Criteria for Depression (the criteria from the Diagnostic and Statistical Manual of Mental Disorders, 5th edition; DSM-5), and the PHQ-2 assesses the two core symptoms of MDD [26]. For patients with HF, several items on the PHQ-9 (abnormal sleep, fatigue and low energy, loss of appetite, poor mood, and lack of concentration) overlap with HF symptoms. The PHQ-2 focuses on loss of interest or pleasure and depressed mood. Although the PHQ-2 has not been established as a tool for making a final diagnosis or for monitoring the severity of depression, it may serve as a useful questionnaire for screening for depressive symptoms in patients with HF [27,28]. If respondents answer "yes" to at least one of the two items on the PHQ-2, the PHQ-9 is administered. If the PHQ-9 score is high or if the item related to suicide is answered with "yes", referral to a psychiatrist is recommended.

Depression and cardiovascular risk

CAD

Depression is associated with poor prognosis in patients with cardiovascular diseases [18] (Fig. 1). There has long been evidence that depression is associated with cardiac prognosis in patients with acute coronary syndromes [29–33]. Lespérance et al. noted that the high BDI scores at admission for myocardial infarction admission, especially in patients with moderate to severe depression, were closely linked to the long-term prognosis regardless of symptom changes [34]. Nakamura et al. reported that depression is a significant risk factor for cardiovascular hospitalization or death after adjusting for cardiac risk factors, even in hospitalized Japanese patients with CAD [35]. A meta-analysis provided further evidence of an association between clinical depression or depressive symptoms and CAD [36]. This association has been confirmed in a variety of populations with CAD, including those with acute coronary syndromes, HF, stable

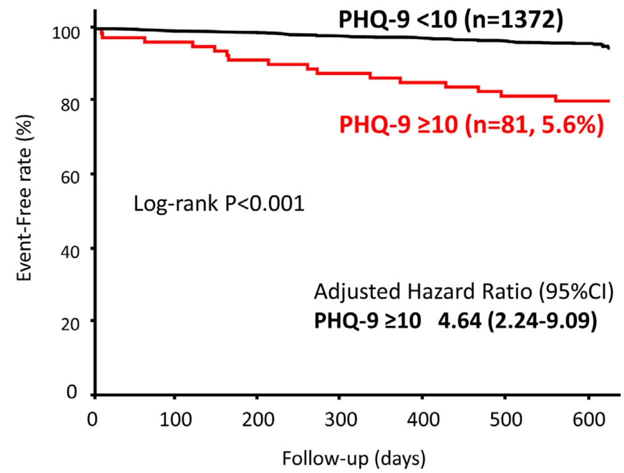

Fig. 1. Cardiovascular outcomes (all-cause mortality or cardiovascular events) in Japanese cardiovascular outpatients with Patient Health Questionnaire (PHQ)-9 scores <10 or PHQ-9 scores ≥10. Modified from Suzuki et al. [18].
CI, confidence interval.

angina, and those who underwent coronary artery bypass surgery. However, individual studies have reported different risk weights and are influenced by other risk factors. Regardless, depression is an independent risk factor for CAD and explains variance unaddressed by conventional risk factors and lifestyle factors. Interestingly, a meta-analysis of prospective observational studies of individuals free of CAD at study enrollment suggests that depression is independently associated with significantly increased risks of CAD and myocardial infarction [37].

HF

Many reports have indicated that depression is an independent risk factor of HF and predicts poor prognosis in patients with HF. In the above-mentioned meta-analysis, the risks of death and cardiovascular events were doubled in HF patients with depression [10]. Kato et al. reported that in a 2-year prospective study of 115 Japanese outpatients with HF, those with depression had higher frequencies of cardiac death and hospitalization for HF than patients without depression [19]. In addition, Suzuki et al. showed that patients with depression had higher frequencies of all-cause mortality and cardiovascular events, including death and hospitalization due to HF, than those without depression [17]. Depression therefore predicts both prognosis and cardiovascular events in patients with HF.

Arrhythmias

A meta-analysis including 17 observational studies reported that depression is associated with a higher risk of sudden cardiac death, ventricular tachycardia/fibrillation, and atrial fibrillation recurrence [38]. The higher risk of serious arrhythmias may, at least partially, explain the higher mortality rate in depressed patients with HF. The Triggers of Ventricular Arrhythmia (TOVA) study suggested that moderate to severe depression predicts the appropriate ICD shock in patients with CAD and an ICD [39]. Furthermore, depressive symptoms have been shown to remain unchanged during ICD treatment lasting over a year or more regardless of shock actuation [40,41]. We previously reported that patients with depression experienced more ICD shocks in the preceding 2 years than those without depression [22]. Risk factors for depression in patients with ICDs include younger age (<50 years), inadequate knowledge of cardiac disease or ICDs, history of psychological problems, lack of social support, higher medical severity, and comorbidities [21]. Previous studies have reported that women generally have a higher incidence of depression than men [42–44]. Thus, sex may also be a risk factor for depression in ICD patients.

Biological relationships: depression and cardiovascular diseases

The biological mechanisms underlying the association between depression and cardiovascular diseases have already been described. However, the relationship between the two conditions is complex, and a clear causal link has not been established. Previous studies have pointed to immune system-mediated inflammation as a common mechanism for both CAD and depression [45]. CAD and many other cardiovascular diseases are mediated by inflammation [46–49]. On the other hand, elevated levels of C-reactive protein (CRP) and interleukin (IL)-6 have also been implicated in depression, suggesting that they are partly related to the development of depression [50]. Recently, a large population-based cohort study in the U.K. examined related inflammatory mechanisms, suggesting that risk factors for CAD, such as levels of triglycerides, CRP, and IL-6, are causally linked with depression [51].

Emotional stress also induces inflammation and cortisol secretion from the adrenal glands [52]. Chronic inflammation promotes atherosclerosis, vascular injury, and depression. Inflammatory markers associated with an increased risk of cardiovascular events, such as CRP, IL-6, and fibrinogen, are also higher in depressed patients with CAD than in nondepressed patients with CAD [53,54]. In addition, stress increases platelet activation [55,56]. Indeed, depressed patients with acute coronary syndrome had higher plasma levels of biomarkers, such as platelet factor 4, beta-thromboglobulin, and platelet/endothelial cell adhesion molecule-1, than nondepressed patients with myocardial infarction or angina [57].

The biological mechanisms underlying HF have been suggested to include increased activity of the sympathetic nervous system, hyperactivity of the hypothalamic-pituitary-adrenal (HPA) axis, and inflammation due to depression [1,2,58,59]. Depression is also associated with disruption of the HPA axis [60]. This axis is the primary neuroendocrine pathway that regulates the stress response and plays a fundamental role in homeostasis; prolonged activation of the HPA axis ultimately results in damage to the cardiovascular system [61,62].

Although the mechanisms underlying the proarrhythmic state are not well understood, there are several potential mechanisms. First, activation of the HPA axis can result in significant changes in structure and function in the cardiovascular system [63,64]. Second, the sympathetic-adreno-medullar (SAM) axis is another important axis that links the sympathetic nervous system and the adrenal medulla. In depressed patients, increased sympathetic tone and decreased parasympathetic tone have been reported, which are observed as decreased heart rate variability and baroreflex sensitivity [65,66]. In particular, this increased sympathetic tone is an important modulator of arrhythmia development. These conditions may contribute to increasing vulnerability to atrial and ventricular arrhythmias.

Social and environmental factors

Social determinants of health include an individual's economic stability, neighborhood and built environment, access to education, access to health care, and social and community relationships. These determinants can be sources of chronic psychosocial stress for individuals suffering from low socioeconomic status, unsafe housing environments, and limited access to health care. Chronic psychosocial stress in turns leads to chronic inflammation through the SAM and HPA axes. All of these inflammatory processes increase cardiovascular risk factors, such as obesity, hypertension, diabetes, and atherosclerosis, and cause increased mortality from cardiovascular diseases [67] (Fig. 2).

Depression is often associated with an unhealthy lifestyle, inactivity, smoking, alcoholism, irritability, obesity, and poor treatment adherence. Depression in HF patients decreases their activity and daily functioning as well as adherence to HF treatment, including medication, diet (salt restriction, fluids), and exercise [6,11,68,69]. In addition, social and family support is often poor in patients with depression. Lack of social support is independently associated with poor cardiovascular prognosis [70]. A longitudinal study examined changes in psychosocial status over time in 105 HF patients and showed that depression increased in patients with lower initial social support but not in patients with higher initial social support [71]. Shimizu et al. demonstrated that depression is associated with reduced daily functioning, as assessed by activities of daily living, and that HF patients with persistent limitations [Performance Measure for Activities of Daily Living 8 (PMADL-8) ≥24] had higher HADS-D scores and a higher prevalence of depression [20].

Psychosocial factors are significantly associated with cardiovascular health outcomes through chronic activation of physiological stress responses and systemic inflammation [67,72]. In particular, psychosocial determinants include chronic psychological stress, subjective social status, job stress, adverse childhood experiences, loneliness/social isolation, and depression [67].

Management of depression

A collaborative approach between cardiologists and psychiatrists is recommended for the management of depression in patients with cardiovascular diseases. The National Institute for Health and Clinical

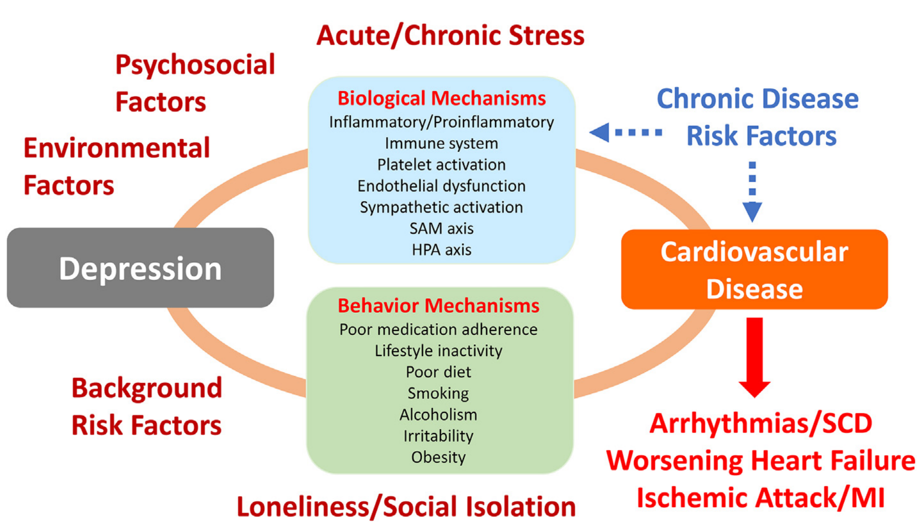

Fig. 2. Common mechanisms of depression and cardiovascular diseases as well as adverse outcome in patients with cardiovascular diseases. HPA, hypothalamic-pituitary-adrenal; MI, myocardial infarction; SAM, sympathetic-adreno-medullar; SCD, sudden cardiac death.

Excellence (NICE) guidelines recommend a "stepped-care model" for managing depression in patients with chronic physical illness [73]. This model consists of four steps: screening for depressive symptoms and intervention to prevent the onset of depression (Step 1), initial (psychotherapy-focused) intervention for mild depression (Step 2), multidisciplinary intervention (including psychiatric pharmacotherapy) for moderate or severe depression that does not respond to the initial intervention (Step 3), and specialized psychiatric treatment, including hospitalization, for severe depression (Step 4). The Japanese Circulation Society (JCS) and Japanese Heart Failure Society (JHFS) 2021 statement on palliative care in cardiovascular diseases outlines a step-by-step approach to care for depression in HF patients in light of Japan's healthcare system [74] (Table 1). Of particular importance to cardiologists and other cardiac health care professionals are Steps 1 and 2 of the stepped-care model. Patients may present with depressive symptoms in various aspects of daily cardiovascular care. It is imperative that healthcare professionals working with patients pay attention not only to physical symptoms but also to psychological status. Step 2, the initial response to the manifestation of depression, begins by determining the specific stressors associated with symptoms of depression and anxiety; possible support to reduce the corresponding stress is discussed [74]. If depression worsens despite this initial response, consultation with a psychiatrist and specialized psychiatric treatment are necessary.

Several reports have demonstrated the benefit of antidepressants for patients with cardiovascular diseases and depression [75,76]. However, the Enhanced Recovery in Coronary Heart Disease (ENRICHD) trial failed to demonstrate a benefit of cognitive-behavioral therapy (CBT) plus a selective serotonin reuptake inhibitor (SSRI) (when indicated), for severe or mild depression in patients with low perceived social support after myocardial infarction [77]. The Sertraline Against Depression and Heart Disease in Chronic Heart Failure (SADHART-CHF) study consisted of 469 patients with chronic HF, a left ventricular ejection fraction ≤45 %, and diagnosed depression; this randomized controlled trial compared sertraline (an SSRI) and matching placebo. The SADHART-CHF study reported no difference in HADS-D scores between the sertraline and placebo groups during the 12-week treatment period and no significant difference in cardiovascular events [78]. However, a subanalysis of these data reported that patients with remission of depression who received either sertraline or the placebo had fewer subsequent cardiovascular events [79]. The effect of antidepressants, such as SSRIs, on depression in ICD patients has not yet been established. Previous studies have suggested that the prevention of ventricular arrhythmia, which requires shock therapy, may be important for improving or preventing depression in patients receiving long-term ICD therapy [80].

Patients with cardiovascular diseases and depression may also benefit from interventions other than pharmacotherapy. Psychotherapy can help depressed patients understand the context that causes depressive symptoms, regain a sense of joy, and develop coping skills [81]. CBT is one such form of psychotherapy that is effective for treating depression in cardiac failure patients but does not improve their prognosis (hospitalization or death) [82]. Kohn et al. reported that CBT conducted preimplantation, predischarge, and at seven routine visits decreased depression by the 9-month follow-up in ICD patients [83]. In another study, CBT also decreased anxiety and ventricular arrhythmias requiring ICD therapy. However, the effects were limited, and the depressive state remained unchanged over the course of the intervention [84].

Cardiac rehabilitation reduces symptoms of depression and improves cardiovascular function [85–87]. However, because depression can also decrease motivation to exercise, the patient and family must be fully informed regarding the importance of adhering to the program [74].

Establishing integrated care systems in the community

The European Society of Cardiology (ESC) and the AHA have released a position paper on depression screening and management [25,88,89]. To date, risk management in cardiovascular diseases has been somewhat effective. However, one risk that remains unaddressed is depression. The management of patients with cardiovascular diseases and depression requires a multifaceted approach, especially integrated management in the community.

The Japanese Cerebrovascular and Cardiovascular Disease Control Act was passed in December 2018 and implemented in December 2019. This national plan describes 10 specific measures to enhance services related to health, medical care, and welfare, including prevention of cerebrovascular and cardiovascular diseases through social collaboration and patient support, the development of a system to provide medical care and rehabilitation for patients with cerebrovascular and cardiovascular diseases, and the provision of appropriate information and consultation support for patients with cerebrovascular and cardiovascular diseases [90]. In the future, community-integrated care for patients with cardiovascular diseases will be established in Japan. This

Table 1
The stepped-care model for managing depression in patients with cardiovascular diseases (modified from the JSC/JHFS 2021 statement on palliative care in cardiovascular disease [74]).

	Target symptoms for intervention	Types of intervention	Professionals for intervention
Step 1	• All symptoms of depression	Screening assessment Supportive response Psychoeducation Active observation Consider referral to a psychiatric professional	Cardiologist Healthcare professionals
Step 2	Mild depression	Mild psychosocial intervention	Certified nurse (e.g., certified nurse in chronic heart failure nursing) or equivalent nurse
	Persistent subthreshold depression[a]	Drug therapy Psycotherapy Consider referral to a psychiatric professional	Cardiologist
Step 3	Moderate to severe disease Mild illness that does not respond to initial intervention	Drug therapy Highly intensive psychotherapy	Psychiatrist Psychiatric nurse (liaison nurse)
	Moderate depression Persistent subthreshold depression[a]	Combination of the above Multidisciplinary intervention Introduction to psychiatric healthcare specialists	Certified psychologist
Step 4	Severe and complicated depression Life crisis High degree of self neglect	Drug therapy Highly intensive psychotherapy Electroconvulsive therapy Crisis intervention Combination of the above Treatment by psychiatric healthcare specialists including hospitalization	Psychiatrist

[a] Subthreshold depression: depressive symptoms without meeting the full criteria of major depression (source: prepared based on NICE guidelines [73]).

system will provide a contact point for consultation between psychiatrists and cardiologists when treating cardiovascular disease patients with the early stages of depression, enabling more accurate assessment and intervention.

Conclusion

Depression is a major risk factor for cardiovascular disease and an important prognostic factor. Although cardiovascular diseases and depression share common pathogenic mechanisms, treatments that improve clinical outcomes in patients with cardiovascular diseases have yet to be developed. Addressing this issue is expected to improve the quality of life in patients with cardiovascular diseases and to improve treatment efficacy and prognosis by increasing adherence to treatment. In the future, studies should identify more detailed targets for treating patients with cardiovascular diseases, collaborate across disciplines (team medicine), and establish a support system in each community.

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Declaration of competing interest

The author has no conflicts of interest to declare.

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